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Ferretting Out Tunneling: An Application to Indian Business Groups

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1 Big picture

This paper studies **tunneling** inside business groups: the transfer of resources from firms where the controlling shareholder has relatively low cash-flow rights to firms where the same controller has relatively high cash-flow rights. The concern is especially relevant in economies where legal protection for minority shareholders is weak and where business groups are common.

- A business group often contains many legally separate, publicly traded firms controlled by the same family or ultimate owner.
- Control and cash-flow rights need not coincide. The controller may control a firm while owning only a small fraction of its cash flows.
- This creates an incentive to move profits away from firms with many outside minority shareholders and toward firms where the controller owns more cash-flow rights.
- The paper's key empirical idea is not to look only at levels of profitability or valuation, but to trace how **earnings shocks propagate across firms within the same group**.

Central question. Can we detect and quantify tunneling by observing how industry-level profit shocks to one group firm show up in the reported profits of other firms in the same business group?

Main answer. Yes. Using Indian business group data, the paper finds that group firms under-respond to their own industry shocks, especially when the controlling family has low cash-flow rights, while other firms in the same group - especially those with high controller ownership - respond positively to those shocks. The evidence points to substantial tunneling, much of it through nonoperating components of profits.

Interpretation. The paper converts an otherwise hidden corporate-governance problem into a testable implication about shock transmission. If money is tunneled, then shocks should not remain where they originate: they should disappear from low-cash-flow-right firms and reappear elsewhere in the group.

2 Incremental contribution of the paper

2.1 Relative to prior corporate-governance evidence

Before this paper, much evidence on tunneling relied on cross-sectional facts such as lower valuation or lower profitability in firms where controlling shareholders had low cash-flow rights. Such facts were suggestive but not decisive, because low valuation could reflect many omitted firm characteristics.

This paper adds a **shock-based methodology**. Instead of asking whether some firms are always less profitable, it asks how reported profits move after plausibly exogenous industry shocks. This makes the test much closer to the mechanism of tunneling.

2.2 Specific incremental contributions

1. **Methodological contribution.** The paper proposes a general empirical method for measuring tunneling based on the propagation of industry earnings shocks across firms in the same business group.
2. **Identification contribution.** It derives a set of asymmetric predictions: money should flow from low-cash-flow-right firms to high-cash-flow-right firms, not symmetrically in both directions.
3. **Measurement contribution.** It applies the method to a large Indian firm panel from CMIE Prowess, using group affiliation, ownership categories, accounting profits, and industry shocks.
4. **Mechanism contribution.** It shows that tunneling appears mainly through **nonoperating profits**, not through operating profits, suggesting mechanisms such as loans, write-offs, miscellaneous items, investment income, and related-party transactions rather than ordinary production or transfer pricing alone.
5. **Market-pricing contribution.** It investigates whether stock prices incorporate tunneling and finds evidence that market valuation partly reflects expected tunneling patterns.

Interpretation. The paper is important because it gives researchers a portable test for an activity that is hard to observe directly. The method is especially useful in settings where detailed internal related-party transactions are unavailable but firm-level accounting data, industry classifications, and ownership proxies exist.

3 Institutional setting and data

3.1 Indian business groups

India is a useful setting because business groups are prominent, many groups are family controlled, and minority-shareholder protection was viewed as imperfect during the sample period. The paper discusses groups such as Tata, Bajaj, Birla, Oberoi, and Mahindra as examples of the broader group structure.

The key institutional feature is that a family can control multiple firms while holding different cash-flow stakes in each firm. This gives the family an incentive to move profits from firms where minority shareholders have larger claims to firms where the family captures more of each rupee.

3.2 Data source

The data come from **Prowess**, a database maintained by the Centre for Monitoring Indian Economy (CMIE). The sample covers annual report information for Indian companies between 1989 and 1999.

- The sample excludes state-owned and foreign-owned firms.
- The full sample contains about 18,600 firm-year observations.
- Group firms account for about 7,521 observations, and stand-alone firms for about 11,079 observations.

- CMIE classifies firms into business groups based on company announcements and qualitative knowledge of group behavior.

3.3 Main performance variable

The main performance measure is

$$\text{PBDIT}_{kt} = \text{profits before depreciation, interest, and taxes.}$$

The authors use accounting profits instead of stock returns because the available stock-price and dividend data are noisy over the sample period.

3.4 Cash-flow-right proxies

The paper uses two proxies for the cash-flow rights of the controlling shareholder:

- **Director equity:** higher director equity is interpreted as a higher controlling-party cash-flow stake.
- **Other ownership:** ownership by small outside shareholders. Higher other ownership is interpreted as lower controlling-party cash-flow rights and more minority-shareholder exposure.

Important measurement caveat. The data mainly capture direct ownership. They do not fully capture indirect ownership through pyramids. This creates measurement error in the ranking of firms by ultimate cash-flow rights. The paper argues that this likely biases results toward zero, because noisy ownership rankings make it harder to detect tunneling.

4 Core empirical idea

4.1 Industry shocks as predicted performance

For firm k in industry I at time t , define return on assets as

$$r_{ktI} = \frac{\text{perf}_{ktI}}{A_{ktI}},$$

where perf_{ktI} is performance and A_{ktI} is assets.

The industry-level return is calculated as an asset-weighted industry average, excluding firm k itself:

$$\hat{r}_{It}^{(-k)} = \frac{\sum_{\ell \neq k} A_{\ell t I} r_{\ell t I}}{\sum_{\ell \neq k} A_{\ell t I}}.$$

The predicted performance of firm k is then

$$\text{pred}_{ktI} = A_{ktI} \times \hat{r}_{It}^{(-k)}.$$

This variable is the paper's proxy for the shock to a firm that would occur if the firm followed the performance of its industry.

Interpretation. The key empirical object is the sensitivity of actual reported profits to predicted profits. If a firm receives a positive industry shock but reported profits rise by less than predicted, some of the shock may have been diverted, dissipated, or hidden. If another group firm responds to this shock despite being in a different industry, that suggests within-group transfer.

4.2 Group shock

For a firm k in group $G(k)$, define the shock to the other firms in the same group as

$$opred_{kt} = \sum_{j \in G(k), j \neq k} pred_{jt}.$$

The paper also decomposes this group shock into shocks to subgroups of firms with high or low controlling-party cash-flow rights:

$Lopred_{kt}$ = shock to low-cash-flow-right firms in the group,

$Hopred_{kt}$ = shock to high-cash-flow-right firms in the group.

5 Predictions of the tunneling hypothesis

The paper derives five main empirical predictions.

1. **Group firms underrespond to their own shocks.** If some earnings are tunneled away, the coefficient on own predicted performance should be lower for group firms than for stand-alone firms.
2. **Low-cash-flow-right firms underrespond more.** The controller has greater incentives to tunnel out of firms where he has lower cash-flow rights.
3. **Group firms respond to other group firms' shocks.** If tunneled money reappears elsewhere in the group, a firm should respond to shocks to other group firms.
4. **Shocks to low-cash-flow-right firms matter more as a source.** More money should be taken from firms where the controller owns less.
5. **High-cash-flow-right firms should receive more.** Money should flow disproportionately toward firms where the controller owns more.

Why asymmetry matters. Simple risk-sharing or coinsurance would predict more symmetric transfers across group firms. Tunneling predicts direction: from low controller cash-flow rights to high controller cash-flow rights. This directional prediction is the empirical core of the paper.

6 Empirical specifications

6.1 Group versus stand-alone firms

The first test compares own-shock sensitivity for group and stand-alone firms:

$$perf_{kt} = \alpha + \beta pred_{kt} + \gamma(Group_k \times pred_{kt}) + \delta Controls_{kt} + \mu_k + \tau_t + \varepsilon_{kt}.$$

The tunneling prediction is

$$\gamma < 0.$$

6.2 Ownership heterogeneity within groups

Within business groups, the paper estimates

$$perf_{kt} = \alpha + \beta pred_{kt} + \gamma(CashRights_k \times pred_{kt}) + \delta Controls_{kt} + \mu_k + \tau_t + \varepsilon_{kt}.$$

Predictions:

- Using director equity, $\gamma > 0$.
- Using other ownership, $\gamma < 0$.

6.3 Response to group shocks

The next specification tests whether firms respond to shocks to other firms in the same group:

$$perf_{kt} = \alpha + \beta pred_{kt} + \gamma opred_{kt} + \delta Controls_{kt} + \mu_k + \tau_t + \varepsilon_{kt}.$$

The tunneling prediction is

$$\gamma > 0.$$

6.4 Source of tunneled funds

To test whether shocks to low-cash-flow-right firms are more likely to be redistributed, the paper estimates

$$perf_{kt} = \alpha + \beta pred_{kt} + \gamma_L Lopred_{kt} + \gamma_H Hopred_{kt} + \delta Controls_{kt} + \mu_k + \tau_t + \varepsilon_{kt}.$$

The tunneling prediction is

$$\gamma_L > \gamma_H.$$

Interpretation. The sequence of specifications moves from broad evidence to more direct mechanism evidence. The first tests establish dissipation from group firms; the later tests show that the dissipation follows the direction predicted by ownership incentives.

7 Identification and empirical design

7.1 Source of identifying variation

The main source of variation is the year-to-year performance of industries. A firm's own industry shock should affect its natural profitability, but it should not mechanically affect unrelated firms in the same group unless there is a within-group transfer mechanism.

Firm fixed effects absorb permanent differences across firms, such as average profitability, group quality, technology, and accounting conventions. Year fixed effects absorb common aggregate shocks.

7.2 Threats and responses

- **Industry misclassification or diversification.** If group firms are more diversified, industry shocks may be measured with more error. The authors check product-level diversification and find that it does not explain the results.
- **Internal capital markets or coinsurance.** Groups may move money for efficiency or insurance reasons. But symmetric coinsurance cannot easily explain why flows follow cash-flow-right incentives.
- **Cross-ownership and dividends.** Mechanical dividend flows might produce within-group correlation. The authors argue the magnitudes are too small and results are robust to excluding dividend income.
- **Ownership measurement error.** Indirect ownership is not fully measured. This likely attenuates the estimates rather than creating the directional results.

Interpretation. The design is not a randomized experiment, but the pattern of asymmetric shock propagation is hard to reconcile with simpler alternatives. The strongest evidence comes from the combination of underresponse by low-cash-flow-right firms and positive responses by other group firms.

8 Tables and figures

The original tables below are extracted directly from the uploaded paper. Closely related tables are placed in pairs on the same row; the final single table is displayed at 0.6 width.

8.1 Table I and Table II: sample facts and first tunneling test

TABLE I
SUMMARY STATISTICS

Sample:	All	Groups	Stand-alones
Total assets	131.80 (525.91)	252.76 (741.6)	49.69 (272.66)
Total sales	94.39 (305.66)	188.16 (459.77)	30.73 (57.84)
Profit before depreciation, interest, and taxes	16.84 (63.84)	32.90 (90.99)	5.94 (30.48)
Ratio of PBDIT to total assets	.126 (.128)	.142 (.115)	.115 (.134)
Ratio of operating profit to total assets	.284 (.285)	.328 (.312)	.254 (.261)
Ratio of nonoperating profit to total assets	-.157 (.259)	-.186 (.288)	-1.38 (.235)
<i>q</i> ratio	.537 (.818)	.645 (.916)	.447 (.714)
Year of incorporation	1974.55 (20.03)	1967.51 (22.89)	1979.33 (16.18)
Director equity	16.70 (18.33)	7.45 (13.05)	22.99 (18.72)
Other ownership	29.90 (17.39)	27.57 (16.06)	31.48 (18.07)
Director equity spread	—	15.19 (14.88)	—
Other ownership spread	—	33.31 (21.66)	—
Sample size	18600	7521	11079

a. Data Source: Prowess, Centre for Monitoring Indian Economy (CMIE), for the years 1989–1999. All monetary variables are expressed in 1995 Rs. crore, where crore represents 10 million.

b. Standard deviations are in parentheses.

c. "Operating profit" refers to manufacturing sales revenue minus total raw material expenses, energy expenses, and wages and salaries. "*q* ratio" is the ratio of market valuation to total assets. "Director equity spread" is the difference between the minimum and maximum level of director equity in a group. "Other ownership spread" is the difference between the minimum and maximum level of other ownership in a group. Ownership and ownership spread variables are measured in percentages and so range from 0 to 100.

TABLE II
SENSITIVITY TO OWN SHOCK: GROUP VERSUS STAND-ALONE
DEPENDENT VARIABLE: PROFIT BEFORE DIT

	(1)	(2)	(3)	(4)
Own shock	1.05 (.02)	.10 (.05)	-4.58 (.48)	-5.10 (.47)
Own shock*	-.30	-.30	-.26	-.27
group	(.02)	(.02)	(.02)	(.02)
Ln assets	.16 (.32)	2.98 (.34)	-.33 (.33)	2.47 (.34)
Own shock* ln assets	—	.10 (.00)	—	1.0 (.01)
Own shock* year of incorp.	—	—	.003 (.000)	.003 (.000)
Sample size	18600	18600	18588	18588
Adjusted R ²	.93	.93	.93	.93

a. Data Source: Prowess, Centre for Monitoring Indian Economy, for years 1989–1999. All monetary variables are expressed in 1995 Rs. crore, where crore represents 10 million. Sample includes both stand-alone and group firms.

b. All regressions also include year fixed effect and firm fixed effects.

c. Standard errors are in parentheses.

Table I. Summary statistics

Table II. Sensitivity to own shock: group versus stand-alone

Read. Table I shows that group firms are much larger and older than stand-alone firms. Group firms also have higher average profits and higher market valuation, but much lower director equity. The average within-group spread in director equity and other ownership is large, which provides useful variation for ranking firms within a group.

Table II provides the first test of tunneling. Stand-alone firms respond almost one-for-one to their own industry shocks. Group firms respond less: the interaction between own shock and group membership is negative, around -0.30 , and robust to controls for size and age.

Economic message. The first table establishes that groups are economically important and ownership stakes vary within groups. The second table shows the first sign of leakage: one rupee of predicted performance translates into less than one rupee of reported performance in group firms.

8.2 Table III and Table IV: ownership gradient and within-group propagation

TABLE III
SENSITIVITY TO OWN SHOCK BY DIRECTOR AND OTHER OWNERSHIP
DEPENDENT VARIABLE: PROFIT BEFORE DIT

Panel A: Director equity				
	Sample:			
	Groups (1)	Groups (2)	Stand- alones (3)	Stand- alones (4)
Own shock	.713 (.009)	-5.075 (.742)	1.058 (.006)	-4.316 (.518)
Own shock * director equity	.025 (.003)	.030 (.003)	.004 (.001)	.019 (.001)
Ln assets	.052 (.733)	4.261 (.807)	-.590 (.176)	1.568 (.178)
Own shock * ln assets	—	.118 (.008)	—	.201 (.006)
Own shock * year of incorp.	—	.002 (.000)	—	.002 (.000)
Sample size	7521	7510	11079	11078
Adjusted R ²	.92	.93	.95	.96

Panel B: Other ownership				
	Sample:			
	Groups (1)	Groups (2)	Stand- alones (3)	Stand- alones (4)
Own shock	.919 (.023)	-5.764 (.743)	1.033 (.052)	-3.983 (.603)
Own shock * other ownership	-.007 (.001)	-.007 (.001)	.001 (.000)	.002 (.000)
Ln assets	1.616 (.724)	5.189 (.806)	-.292 (.166)	2.049 (.180)
Own shock * ln assets	—	.103 (.008)	—	.154 (.006)
Own shock * year of incorp.	—	.003 (.003)	—	.002 (.000)
Sample size	7521	7510	11079	11078
Adjusted R ²	.92	.93	.95	.96

a. Data Source: Prowess, Centre for Monitoring Indian Economy, for years 1989–1999. All monetary variables are expressed in 1995 Rs. crore, where crore represents 10 million.
b. All regressions also include year fixed effect and firm fixed effects.
c. Standard errors are in parentheses.

TABLE IV
SENSITIVITY OF GROUP FIRMS TO GROUP AND SUBGROUP SHOCKS
DEPENDENT VARIABLE: PROFIT BEFORE DIT

	(1)	(2)	(3)	(4)	(5)
Own shock	.730 (.009)	.732 (.009)	.732 (.009)	.732 (.009)	.732 (.009)
Group shock	.011 (.001)	—	—	—	—
Shock below median (director equity)	—	.016 (.002)	—	—	—
Shock above median (director equity)	—	-.002 (.005)	—	—	—
Shock below 66th ptile (director equity)	—	—	.015 (.002)	—	—
Shock above 66th ptile (director equity)	—	—	-.001 (.001)	—	—
Shock above median (other ownership)	—	—	—	.014 (.002)	—
Shock below median (other ownership)	—	—	—	.007 (.004)	—
Shock above 33rd ptile (other ownership)	—	—	—	—	.017 (.002)
Shock below 33rd ptile (other ownership)	—	—	—	—	-.002 (.004)
Sample size	7521	7521	7521	7521	7521
Adjusted R ²	.93	.92	.92	.92	.92

a. Data Source: Prowess, Centre for Monitoring Indian Economy, for years 1989–1999. All monetary variables are expressed in 1995 Rs. crore, where crore represents 10 million.
b. Sample is group firms only.
c. “Shock below median (director equity)” is a variable that sums the industry shocks to all the firms in the same group (excluding the firm itself) that have below median level of director ownership in their group. All the other subgroup shocks are defined accordingly.
d. Also included in each regression are the logarithm of total assets, year fixed effects, and firm fixed effects.
e. Standard errors are in parentheses.

Table III. Sensitivity to own shock by director and other ownership

Table IV. Sensitivity of group firms to group and subgroup shocks

Read. Table III shows that, among group firms, sensitivity to own industry shocks increases with director equity and decreases with other ownership. In other words, firms where the controlling party has higher cash-flow rights keep more of their own shocks. The same relationships are much weaker or absent among stand-alone firms.

Table IV shows that group firms respond to shocks to other firms in the same group. The coefficient on the overall group shock is positive. When the group shock is split by ownership, the response is much stronger for shocks originating in low-cash-flow-right firms, exactly where tunneling incentives are strongest.

Economic message. These two tables move the evidence from “group firms look different” to “group firms behave in the direction predicted by tunneling.” Low-cash-flow-right firms appear to be the source of transfers; other group firms receive part of the shock.

8.3 Table V and Table VI: destination of funds and accounting channel

TABLE V
SENSITIVITY TO GROUP SHOCK BY LEVEL OF DIRECTOR OWNERSHIP IN GROUP
DEPENDENT VARIABLE: PROFIT BEFORE DIT

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Level in group:	Lower ½	Top ½	Below topmost firm		Topmost firm			
Own shock	.62 (.01)	.89 (.02)	.63 (.01)	.63 (.01)	.63 (.01)	1.01 (.02)	1.01 (.02)	1.01 (.02)
Group shock	.013 (.002)	.010 (.002)	.012 (.001)	—	—	.020 (.008)	—	—
Shock below 66th ptile (director equity)	—	—	—	.015 (.002)	—	—	.032 (.012)	—
Shock above 66th ptile (director equity)	—	—	—	.003 (.006)	—	—	.007 (.018)	—
Shock below 33rd ptile (other ownership)	—	—	—	—	-.000 (.004)	—	—	-.013 (.025)
Shock above 33rd ptile (other ownership)	—	—	—	—	.017 (.002)	—	—	.034 (.011)
Sample size	4905	2616	5780	5780	5780	1741	1741	1741
Adjusted R ²	.90	.95	.90	.97	.97	.97	.97	.97

a. Data Source: Prowess, Centre for Monitoring Indian Economy, for years 1989–1999. All monetary variables are expressed in 1995 Rs. crore, where crore represents 10 million.
b. Firms are separated into different “Level in group” based on their within-group level of director equity. For example, “Topmost Firm” are the set of firms that have the highest level of director ownership in their group.
c. Also included in each regression are the logarithm of total assets, year fixed effects, and firm fixed effects.
d. Standard errors are in parentheses.

Table V. Sensitivity to group shock by level of director ownership in group

TABLE VI
SHOCK SENSITIVITY: AN ACCOUNTING DECOMPOSITION

Panel A: Sensitivity to own shock		
Sample:	Groups	Stand-alones
Dep. variable:		
Operating profits	1.22 (.018)	1.17 (.009)
Nonoperating profits	-.478 (.014)	-.103 (.006)
Panel B: Sensitivity to own shock by director ownership		
Sample:	Groups	Stand-alones
Dep. variable:		
Operating profits	.0123 (.0056)	.0082 (.0013)
Nonoperating profits	.0131 (.0043)	-0.0038 (.0008)
Panel C: Sensitivity to group shock by level of director ownership in group		
Sample:	Topmost firm	Below topmost firm
Dep. variable:		
Operating profits	.0066 (.0128)	.0114 (.0026)
Nonoperating profits	.0134 (.0078)	.0006 (.0020)

a. Data Source: Prowess, Centre for Monitoring Indian Economy, for years 1989–1999. All monetary variables are expressed in 1995 Rs. crore, where crore represents 10 million.
b. Each coefficient contains the result of a separate regression in which the dependent variable is either operating profits or nonoperating profits, as indicated. In Panel A the reported coefficient is the coefficient on “Own shock.” In Panel B the reported coefficient is the coefficient on “Own shock · director equity.” In Panel C, the reported coefficient is the coefficient on “Group Shock.” Also indicated in each regression are the logarithm of total assets, year fixed effects, firm fixed effects, and “Own shock” (Panels B and C).
c. In Panel C the subsamples are for group firms only. Topmost firm and below topmost firms are defined using director’s equity. For example, “Topmost firm” are the set of firms that have the highest level of director ownership in their group.
d. Standard errors are in parentheses.

Table VI. Shock sensitivity: an accounting decomposition

Read. Table V asks whether money goes to the top of the ownership structure. The strongest patterns appear when comparing the firms with the highest director ownership in their group to the remaining firms. Topmost firms are more sensitive to group shocks, especially shocks to firms with lower director equity or higher outside ownership. Some estimates are noisy, but the direction is consistent with tunneling toward high-cash-flow-right firms.

Table VI decomposes profits into operating and nonoperating components. The evidence of tunneling is concentrated in nonoperating profits. Operating profits do not show the same tunneling pattern. Nonoperating losses and gains appear to offset ordinary operating shocks and move resources across group firms.

Economic message. Table V addresses the destination of funds; Table VI addresses the mechanism. The paper’s mechanism evidence suggests that tunneling in Indian groups is not primarily hidden in ordinary production margins. It appears in accounting categories where related-party loans, write-offs, investment income, and extraordinary items can be recorded.

8.4 Table VII: market valuation and expected tunneling

TABLE VII
SENSITIVITY TO OWN AND GROUP SHOCK BY FIRM AND GROUP Q RATIOS
DEPENDENT VARIABLE: PROFIT BEFORE DIT

	(1)	(2)	(3)	(4)
Own shock	-.046 (.056)	.388 (.027)	.600 (.017)	.049 (.060)
Own shock * firm Q	.178 (.013)	—	—	.143 (.016)
Own shock * relative Q	—	.143 (.011)	—	—
Own shock * group Q	—	—	.414 (.037)	.171 (.044)
Group shock	-.008 (.003)	.010 (.002)	.011 (.003)	-.008 (.004)
Group shock * firm Q	.012 (.001)	—	—	.012 (.001)
Group shock * relative Q	—	.008 (.001)	—	—
Group shock * group Q	—	—	.006 (.007)	-.001 (.006)
Adjusted R^2	.94	.94	.93	.94

a. a. Data Source: Prowess, Centre for Monitoring Indian Economy, for years 1989–1999. All monetary variables are expressed in 1995 Rs. crore, where crore represents 10 million.

b. Sample is group firms only.

c. "Firm Q" is a variable that represents the estimated firm fixed effects in a regression of firm-level q ratios (market valuation over total assets) on log(total assets), year fixed effects, industry fixed effects, and firm fixed effects. "Group Q" is a variable that represents the estimated group fixed effects in a regression of firm-level q ratios on log(total assets), year fixed effects, industry fixed effects, and group fixed effects. "Relative Q" is the difference between "Firm Q" and the mean of "Firm Q" within groups.

d. Also included in each regression are the logarithm of total assets, year fixed effects, and firm fixed effects.

e. Standard errors are in parentheses.

Table VII. Sensitivity to own and group shock by firm and group Q ratios

Read. Table VII examines whether market valuation reflects tunneling patterns. Firms with higher own q are more sensitive to their own shocks and to group shocks. Relative q shows a similar pattern. Group q is positively related to sensitivity to own shocks, suggesting that groups that tunnel less away from firms are valued more highly.

Economic message. The market appears to price at least part of the tunneling environment. Firms expected to lose less to tunneling, or to receive more through the group, have higher valuations. This does not imply that market pricing is perfect, but it indicates that investors are not entirely blind to group-level expropriation incentives.

9 Main results

9.1 Evidence of tunneling

The paper finds a consistent set of results:

- Group firms respond less to their own industry shocks than stand-alone firms.
- Within groups, firms with higher director ownership retain more of their own shocks.
- Within groups, firms with higher other ownership retain less of their own shocks.

- Group firms respond positively to shocks affecting other firms in the same group.
- Group firms respond more to shocks originating in low-cash-flow-right firms.
- Firms at the top of the within-group ownership ranking appear to benefit more from group shocks.

9.2 Magnitude

A central magnitude is that more than roughly one-quarter of the marginal rupee of profits in low-cash-flow-right group firms appears to be dissipated or diverted. The group-shock estimates imply that a sizable share of the missing money reappears elsewhere in the group, though not all of it is observed in the sample.

9.3 Mechanism

The accounting decomposition is central. Tunneling appears in nonoperating profits rather than operating profits. This is consistent with channels such as:

- loans across group firms at nonmarket terms,
- asset sales or purchases at distorted prices,
- write-offs and extraordinary items,
- investment income and miscellaneous financial transactions,
- other related-party transactions outside ordinary production accounts.

10 Interpretation and economic intuition

The paper's findings fit a simple agency problem. The controlling family maximizes the value of its own cash-flow claims, not the value of all firms equally. When a low-cash-flow-right firm receives a good shock, the family captures only a small fraction if the money stays there. If the family can move the money to a high-cash-flow-right firm, it captures a larger fraction. Therefore, good shocks to low-cash-flow-right firms should be partly removed from those firms and partly show up in other group firms.

This logic also explains why the direction of flows matters. A pure insurance mechanism would move resources toward firms hit by negative shocks, regardless of cash-flow-right rankings. Tunneling, however, is motivated by ownership incentives and should move money toward the controller's stronger cash-flow claims.

11 Limitations and caveats

- **Ownership is imperfectly measured.** The authors cannot fully observe indirect cash-flow rights through pyramids, so ownership rankings are noisy.
- **The estimates are accounting-based.** The paper relies on accounting profits, not stock returns, because return data are noisy.

- **Groups may have benefits too.** The paper is not a full welfare evaluation of business groups. Groups may also provide reputation, internal capital markets, political connections, or transaction-cost savings.
- **The evidence is not a direct audit trail.** The paper detects tunneling through patterns in shock propagation, not through individual related-party transactions.
- **Some estimates are noisy.** The destination-of-funds results in Table V are suggestive but less statistically sharp than the own-shock and subgroup-shock tests.

Interpretation. The caveats do not overturn the main contribution. Because tunneling is hidden by nature, a clean audit trail is often unavailable. The paper’s contribution is to infer tunneling from a systematic pattern that follows the economics of control and cash-flow rights.

12 Short summary

- **Question:** Do Indian business groups tunnel profits across listed firms?
- **Method:** Use industry profit shocks to trace whether shocks disappear from one group firm and reappear in other group firms.
- **Key prediction:** Shocks should flow from low-cash-flow-right firms to high-cash-flow-right firms.
- **Data:** CMIE Prowess panel of Indian firms, 1989-1999, with group affiliation, accounting data, and ownership categories.
- **Main finding:** Group firms underrespond to their own shocks; within groups, the pattern of underresponse and cross-response follows ownership incentives.
- **Mechanism:** Tunneling is concentrated in nonoperating profits.
- **Market implication:** Market valuations partly reflect expected tunneling patterns.

13 Conclusion

Bertrand, Mehta, and Mullainathan provide a powerful empirical strategy for detecting tunneling in business groups. The paper’s core insight is that hidden transfers leave a visible statistical footprint: profits should not simply follow a firm’s own industry shocks, but should also reflect the shocks to other firms in the same group in a way that depends on the controlling shareholder’s cash-flow rights.

The Indian evidence supports this prediction. Low-cash-flow-right group firms appear to lose a significant share of marginal profits, while other firms in the group, especially those more valuable to the controller, benefit. The accounting decomposition points toward nonoperating profit items as the main channel. The broader lesson is that weak corporate governance can distort the meaning of accounting profits, minority-shareholder returns, and firm valuation inside business groups.